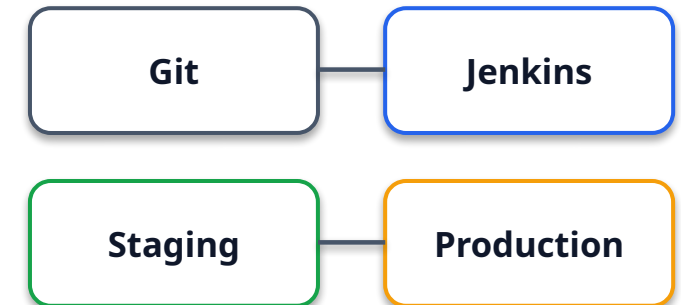


KijaniKiosk Capstone

Track A - Infrastructure-first deployment pipeline

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17 August 2026



The Problem

KijaniKiosk needs to deploy the kk-payments microservice to a shared Kubernetes cluster while keeping staging and production isolated, ensuring deployments are auditable through Git, and catching failures before they reach users. Today, staging health depends on manual checks, with no automated validation gate preventing an unhealthy release from being promoted to production.

Operational gap

Manual staging validation

Risk

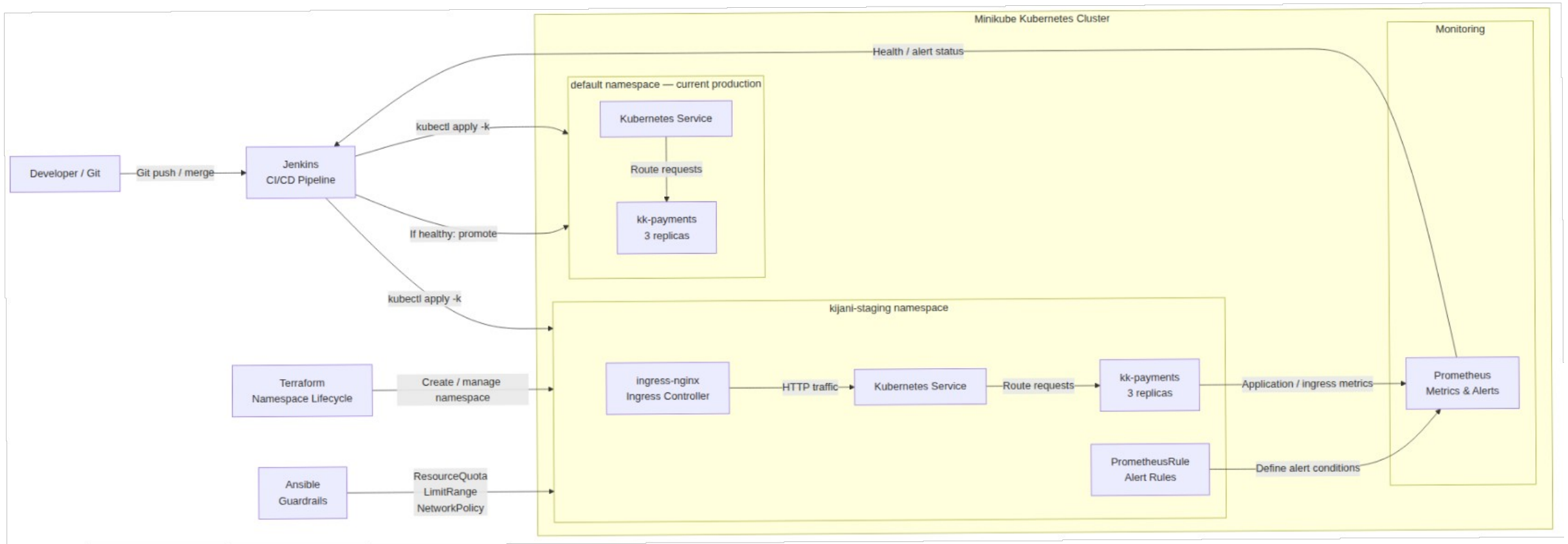
Unhealthy release reaches users

Goal

Automated health-gated promotion

Architecture

Every connection represents a real deployment, control, or observability flow.



Key Decision

Kustomize overlays over Helm charts for environment separation

CHOSEN: Kustomize

Base manifests + staging/production overlays; ConfigMapGenerator for environment-specific values.

NOT CHOSEN: Helm

Helm charts with per-environment values files.

WHY

Native to kubectl, readable plain YAML, and the staging/production diff stays immediately visible.

Trade-off

Overlay sprawl becomes harder to maintain as environments grow beyond roughly 2-3.

Production change

If environment count or configuration complexity grows significantly, evaluate Helm or another templating/package approach.

AI Tooling

AI assistance was treated as generated infrastructure code, not as an authority.

What AI got right

Generated the Kustomize overlay structure, namespace overrides, standard app.kubernetes.io labels, and /health readiness/liveness probes.

What AI got wrong

Generated an Ingress that depended on the nginx admission webhook without documenting or validating the prerequisite service.

What I changed

Added a pipeline prerequisite-verification stage for the PrometheusRule, ingress-nginx service, and Prometheus service before deployment.

Governance lesson

The generated configuration was run and validated before application; the missing admission-service dependency was caught through deployment testing.

Production Gaps

Specific changes required before real KijaniKiosk customer traffic

1. Production isolation

GAP

Production currently runs in the default namespace.

Solution

Create kijani-production with Terraform; apply the same Ansible ResourceQuota, LimitRange, and NetworkPolicy guardrails; update the production Kustomize overlay and Jenkins PRODUCTION_NAMESPACE.

2. Production monitoring

GAP

Production lacks a production-specific PrometheusRule and observability resources in its Kustomization.

Solution

Add production PrometheusRule resources and include them in k8s/overlays/production/.

3. Real-cluster readiness

GAP

The current foundation is validated on Minikube rather than a production Kubernetes environment.

Solution

Validate ingress, monitoring, storage/network behavior, capacity, and rollout/rollback behavior on the target production cluster.